C

CAMBRIDGE ADDITIONAL MATHEMATICS (0606) • UNIT 8.01

Integration & Kinematics

PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Integration & Kinematics Practice Exercises

1 MCQ

Evaluate the definite integral: $\int_0^1 (2x+1)^3\,dx$.

A $\frac{80}{2}$

B $\frac{84}{8}$

C $\frac{80}{4}$

D $\frac{80}{8}$

2 MCQ

Evaluate the definite integral: $\int_0^1 (3x+1)^3\,dx$.

A $\frac{255}{4}$

B $\frac{255}{3}$

C $\frac{259}{12}$

D $\frac{255}{12}$

3 MCQ

Evaluate the definite integral: $\int_0^1 (4x+1)^3\,dx$.

A $\frac{625}{16}$

B $\frac{624}{4}$

C $\frac{628}{16}$

D $\frac{624}{16}$

4 MCQ

Evaluate the definite integral: $\int_0^1 (5x+1)^3\,dx$.

A $\frac{1295}{5}$

B $\frac{1295}{4}$

C $\frac{1295}{20}$

D $\frac{1299}{20}$

5 MCQ

Evaluate the definite integral: $\int_0^1 (6x+1)^3\,dx$.

A $\frac{2400}{6}$

B $\frac{2400}{24}$

C $\frac{2404}{24}$

D $\frac{2400}{4}$

6 MCQ

Evaluate the definite integral: $\int_0^1 (7x+1)^3\,dx$.

A $\frac{4095}{4}$

B $\frac{4099}{28}$

C $\frac{4095}{28}$

D $\frac{4095}{7}$

7 MCQ

Evaluate the definite integral: $\int_0^1 (8x+1)^3\,dx$.

A $\frac{6560}{4}$

B $\frac{6560}{8}$

C $\frac{6564}{32}$

D $\frac{6560}{32}$

8 MCQ

Evaluate the definite integral: $\int_0^1 (9x+1)^3\,dx$.

A $\frac{9999}{4}$

B $\frac{9999}{9}$

C $\frac{10003}{36}$

D $\frac{9999}{36}$

9 MCQ

The diagram shows part of the curve $y = 6x - x^2$, which intersects the x-axis at $(0, 0)$ and $(6, 0)$.
Find the area of the region enclosed between the curve and the x-axis.



A 36

B 27

C 54

D 72

10 MCQ

A particle moves in a straight line such that its velocity, v m s^{-1}, at time t seconds is given by $v = 6t^2 - 18t + k$, where k is a constant.\n\nGiven that the particle is momentarily at rest when $t = 2$, find the total distance traveled by the particle in the first 2 seconds.

A 6 m

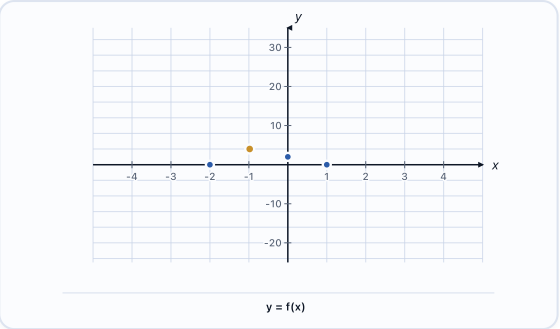
B 4 m

C 5 m

D 8 m

11 MCQ

The curve C has equation $y = x^3 - 3x + 2$. The tangent to C at the point $P(2, 4)$ intersects the curve again at point Q. Find the area of the finite region completely enclosed between the curve C and the tangent line.



A 84

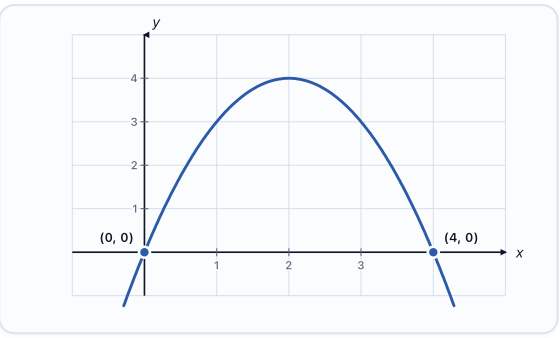
B 64

C 12

D 108

12 MCQ

The diagram shows the curve $y = 4x - x^2$. Find the area of the region enclosed by the curve and the x-axis.



A 16

B $\frac{64}{3}$

C $\frac{16}{3}$

D $\frac{32}{3}$

13 MCQ

A particle moves in a straight line such that its acceleration, a m s^{-2}, at time t seconds is given by $a = 6t - 18$. The particle is at instantaneous rest when $t = 1$. Given that the displacement of the particle from a fixed point O is $s = -10$ m when $t = 2$, find the displacement of the particle, in metres, when it is next at instantaneous rest.

A -25

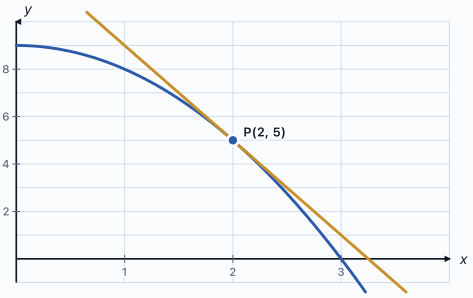
B -13

C -37

D -5

14 MCQ

The diagram shows part of the curve $y = 9 - x^2$ for $x \geq 0$ and the tangent to the curve at the point $P(2, 5)$. Find the exact area of the finite region bounded by the curve, the tangent at P, and the x-axis.



A $\frac{125}{192}$

B $\frac{11}{24}$

C $\frac{8}{3}$

D $\frac{25}{8}$

15 MCQ

A particle moves in a straight line such that its acceleration, a m s^{-2}, at time t seconds is given by $a = 6t - 4$. When $t = 1$, the velocity of the particle is 5 m s^{-1}. Find the velocity of the particle when $t = 3$.

A 19 m s^{-1}

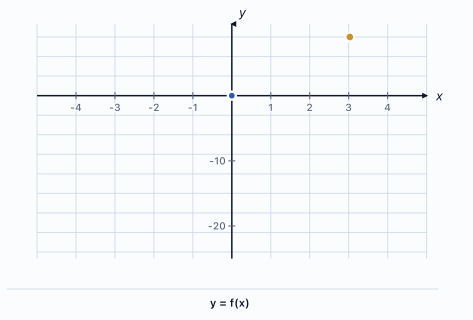
B 27 m s^{-1}

C 21 m s^{-1}

D 15 m s^{-1}

16 MCQ

A curve has equation $y = 6x - x^2$. The straight line $y = mx$, where $0 < m < 6$, intersects the curve at the origin and at the point P. Given that the area of the region enclosed between the curve and the line is $\frac{32}{3}$, find the value of the constant m.



A 4

B 2

C 3

D 1

17 MCQ

Calculate the solution set for the exponential equation $2^{x+4} = 5^x$, giving x in terms of common logarithms.

A $x = \frac{4 \ln 2}{\ln 5 - \ln 2}$

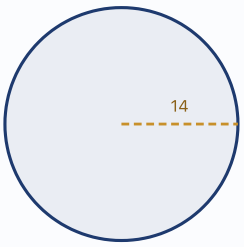
B $x = \frac{\ln 2}{4(\ln 5 - \ln 2)}$

C $x = \frac{4 \ln 5}{\ln 5 - \ln 2}$

D $x = \frac{4}{\ln 5 - \ln 2}$

18 MCQ

Suppose a sector of a circle of radius $r = 14$ cm has central angle $\theta = 1.2$ rad. What is the area of the sector?



A $16.8\ \text{cm}^2$

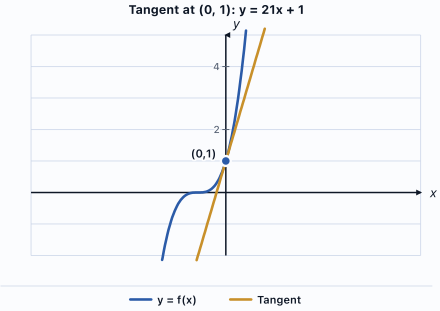
B $235.2\ \text{cm}^2$

C $117.6\ \text{cm}^2$

D $122.6\ \text{cm}^2$

19 MCQ

Identify the equation of the tangent line to the curve $y = (7x + 1)^3$ at the point where $x = 0$.



A $y = 21x + 1$

B $y = 21x + 3$

C $y = 21x - 1$

D $y = 7x + 1$

20 MCQ

Determine the numerical value of the definite integral: $\int_0^1 (3x + 1)^3\, dx$.

A $\frac{255}{12}$

B $\frac{255}{4}$

C $\frac{255}{3}$

D $\frac{259}{12}$

21 MCQ

Calculate the simplified result of the definite integral: $\int_0^1 (4x + 1)^3\, dx$.

A $\frac{625}{16}$

B $\frac{624}{4}$

C $\frac{628}{16}$

D $\frac{624}{16}$

22 MCQ

Find the exact value of the expression the definite integral: $\int_0^1 (5x+1)^3\,dx$.

A $\frac{1295}{20}$

B $\frac{1295}{5}$

C $\frac{1295}{4}$

D $\frac{1299}{20}$

23 MCQ

Work out the exact value of the definite integral: $\int_0^1 (6x+1)^3\,dx$.

A $\frac{2404}{24}$

B $\frac{2400}{4}$

C $\frac{2400}{24}$

D $\frac{2400}{6}$

24 MCQ

Simplify and evaluate the definite integral: $\int_0^1 (7x+1)^3\,dx$.

A $\frac{4095}{4}$

B $\frac{4099}{28}$

C $\frac{4095}{7}$

D $\frac{4095}{28}$

25 MCQ

Ascertain the exact result of the definite integral: $\int_0^1 (8x+1)^3\,dx$.

A $\frac{6560}{32}$

B $\frac{6560}{4}$

C $\frac{6560}{8}$

D $\frac{6564}{32}$